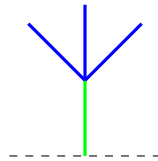


Quiz Week 2

March 13, 2013

1. *Left goes first. Who wins $A+B$?*

Proof. We have the game B which is a nim heap of size 3 and game A which is



If Left goes first, Left has three options

- 1) Left chops the bottom green, Right takes the 3 coins and Right wins
- 2) Left chops any of the top blue pieces, Right takes 1 coin and they repeat this until either all the coins are gone in which case Left chops the green piece, or Right chops the green, and Left takes the remaining coin, so Left wins
- 3) Left takes one or more coins. If Left takes 2 coins, Right chops the green or takes the coin and Left wins

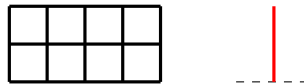
We see that the second and third option is optimal for Left, so that is the one Left will choose. Therefore, the game $A + B$ is Left win if Left goes first.

If Right goes first, Right takes 1 coin, then Left takes 1 blue stem, and this continues until Right takes the last coin in which case Left chops the green stem and wins. Therefore, it is a Left win if Right goes first.

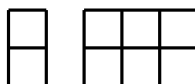
□

2. *Left goes first. Who wins $C+F$?*

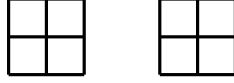
Proof.



Left's only option is to cut the cake into



or



Let us first try Left moving to the first position. Then Right can cut the 1×2 cake, leaving Left to cut the 3×2 cake into a 2×2 and a 1×2 . Right can keep repeating this until Left has no more moves, and since Right also has the red stalk, Right wins.

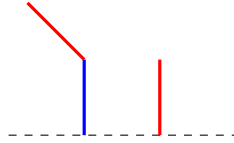
Now, if Left moves to the second position, Right can cut the red stalk. Then there are two 2×2 squares so Right wins (Left cuts one of the squares in half, leaving one move for Left and 3 possible moves for Right).

Therefore, $C + F$ is a Right win if Left goes first.

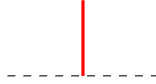
If Right goes first, Right takes the red stem. Then Left cuts the cake into two 2×2 and Left wins. Therefore, Left wins if Right goes first. $\square$

3. Left goes first. Who wins $E + F$?

Proof.



If Left goes first, Left has only one option which is to chop down

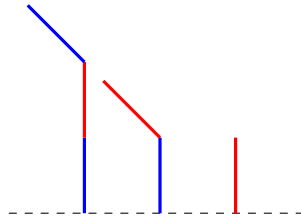


which just leaves a single red stalk. Therefore, Right wins if Left goes first.

If Right goes first, Right can take the Red piece on top of the blue. Then Left takes the blue stem and right takes the last red stem and wins. Therefore, Right wins if Right goes first (Notice that using the values of each of these positions we have $\frac{1}{2} - 1 = -\frac{1}{2}$ which is negative, so Right wins as we determined above). $\square$

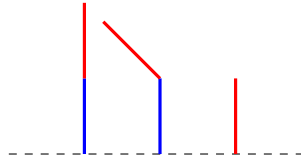
4. Left goes first. Who wins $D + E + F$?

Proof.



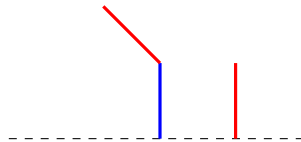
Looking at this, we see that Left has three options:

1) First, say Left chops the top blue piece on game D to get



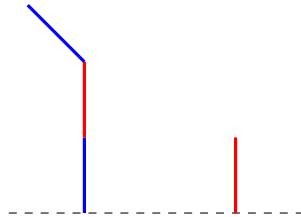
Then, if Right chops one of the red stalks on top of a blue stalk, Left chops the opposite blue stalk and Left wins. If Right chops the single stalk, we have two positive games so Left wins. Thus, this is a Left win game.

2) Now, say Left chops the bottom blue piece on game D to get



Then, if Right chops the red stalk on top of the blue stalk, Right wins, so this position is a Right win.

3) Lastly, say Left chops the blue piece on game E to get

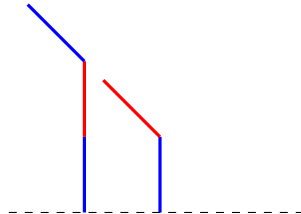


Then, if Right chops the red stalk in the middle of the two red stalks, Right wins.

Therefore, we see that option 1) is best for Left and that this is a Left win game.

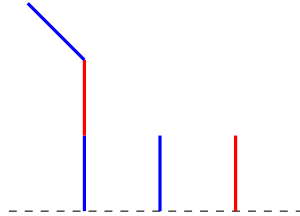
If Right goes first, Right has 3 options:

1) Right moves to



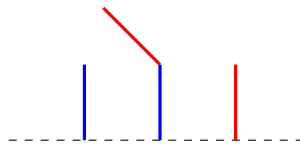
in which case Left can chop any blue piece and win.

2) Right moves to



in which case Left can chop any piece in the first game (the 3 stem one) and win.

3) Right moves to



in which case Left chops the Blue in the middle game (the red stem on top of the blue stem game) and wins.

Therefore, Right loses if Right goes first (Notice using the values of each position we get $\frac{3}{4} + \frac{1}{2} + (-1) = \frac{1}{4}$ which is positive, so Left wins as we determined above). $\square$